Tanzila Rahman (She/Her)

Personnel Website | Linkedin | tanzila.himu@gmail.com

Research Interests

Computer vision. Machine learning, Multimodal learning, Generative AI, Agentic AI, Low precision QAT, Vision and language.

Education

University of British Columbia, Vancouver, BC, Canada — September 2018 - April 2024
Ph.D. in Computer Science | GPA: 88.7% | Advisor: Dr. Leonid Sigal

University of Manitoba, Winnipeg, Manitoba, Canada — September 2016 - July 2018
M.Sc. in Computer Science | GPA: 4.10/4.50 | Advisor: Dr. Yang Wang

Jahangirnagar University, Dhaka, Bangladesh — May 2012-October 2013
M.Sc. in Computer Science and Engineering | 3.91/4.00 (2nd out of 35)

Jahangirnagar University, Dhaka, Bangladesh — April 2007- March 2011
B.Sc. in Computer Science and Engineering | 3.74/4.00 (1st out of 35)

Experience

Huawei Technologies, Vancouver, Canada | Senior Researcher — September 2025 - present
- Lead the research direction and technical roadmap for advanced AI systems.
- Design and optimize low-precision (e.g., FP8/FP4) training and inference techniques for multimodal AI.

Vector Institute, Toronto, Canada | Postdoctoral Fellow — May 2024 - August 2025
- Supervisor: Prof Dr. Leonid Sigal and Prof Dr. Renjie Liao
- Conduct advanced research under the supervision of Professor Leonid Sigal and contribute to scholarly publications, and assist in mentoring students.
- Design and develop end-to-end AI models that maximize data efficiency and deliver robust, scalable performance.

UBC, Vancouver, Canada | Research Assistant — September 2018 - April 2024
- Supervisor: Leonid Sigal
- Generative AI: Focus on generating high-quality images with finer details and better global structure preservation.
- Multimodal Learning: Leverage the complementary nature of different modalities to enhance the understanding, representation, and analysis of complex data.

Snap Research, Toronto, Canada | Collaborator & Research Intern — May 2022 - May 2023
- Developed an autoregressive approach with memory attention that maintains consistency across the frames and published papers in CVPR 2023 and CRV 2025 (Oral).

Amazon GO, Vancouver, Canada | Applied Scientist Intern — May 2020 - September 2020
- Analyzed draw-backs of existing methods and worked to develop an approach to detect the body joints of all the people in all the frames and track each person over the whole video.

Secure Link Service BD Ltd, Bangladesh | Software Engineer — February- January 2015
- Worked on developing web-based applications using asp.net MVC, C#, Javascript, JQuery, and OOP.

HawarIT Software Service Limited, Bangladesh | Software Engineer — March 2011- January 2014
- Analyzed and developed Windows and web applications. Did R&D for CAD/GIS-based web applications using Autodesk MapGuide Studio 2010.

Teaching

Jahangirnagar University, Bangladesh | Lecturer in Dept. of CSE — December 2014-September 2016
- Courses Taught: Medical Imaging (CSE-614), Artificial Intelligence (CSE-451), Web Engineering and Multimedia (CSE-354), Object-Oriented Analysis and Design (CSE-257), Information System Design (CSE-311).
- Responsibilities: Deliver lectures, design course materials and assessments, conduct tutorials, evaluate student performance, and provide academic guidance and mentorship.

University of British Columbia, Vancouver, Canada | Teaching Assistant — September 2019-April 2024

- Courses Taught: Introduction of Software development (CPSC 211), Software Engineering Project (CPSC 319), Advanced Software Engineering (CPSC 410), Computer Vision (CPSC 425), Topics in Artificial Intelligence (CPSC 532S).
- Responsibilities: Design assignments and examinations, conduct tutorial sessions, and evaluate student performance.

Awards and Honors

- Received President's Academic Excellence Initiative Ph.D. Award in May 2020 - 2024.
- Received Amazon Go Fellowship in September 2019.
- Received a research grant from Vector Institute of Artificial Intelligence in 2019-2024.
- Ontario Trillium Scholarship, The Ministry of Training, Colleges and Universities (MTCU), 2018-2022. (Declined).
- University of Manitoba Graduate Fellowship (UMGF), Faculty of Graduate Studies, University of Manitoba, 2018-2022. (Declined)
- Conference Travel Grant, Department of Computer Science and Faculty of Science, University of Manitoba, 2017.
- Manitoba Graduate Scholarship (MGS), Government of Manitoba, 2016-2018 (Government of Manitoba provides funds for Manitoba's best students).
- University of Manitoba Graduate Fellowship (UMGF), 2016-2018.
- International Graduate Student Entrance Scholarship (IGSES), Faculty of Graduate Studies, University of Manitoba, 2016-2017.
- Government talent pool scholarship (For B.Sc. result).

Memberships/Organizer

- Program committee member in AAAI-2025.
- Computer Vision Foundation: Member
- Workshop organizer: [PixFoundation CVPR2025]

Conference Reviewer

- IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR). 2019 – 2026.
- Neural Information Processing Systems (NeurIPS). 2022 – 2026.
- European Conference on Computer Vision (ECCV). 2022, 2024, 2026.
- International Conference on Computer Vision (ICCV). 2021, 2023, 2025.
- Winter Conference on Applications of Computer Vision (WACV). 2020 – 2024.
- AAAI Conference on Artificial Intelligence. 2020 – 2022, 2024.

Mentoring

- Jiayun Luo, Research Topic: Identify high-impact visual tokens to enhance reasoning and information flow in VLM. 2025.
- Wan-Cyuan Fan, Research Topic: Develop an Agentic AI based framework for automated model and metric routing. 2024.
- Shivansh Rao, Research Topic: Video based person re-identification. 2017.
- Taimur Islam, Ataur Rahman Bappy, Research Topic: Filtering political sentiment in social media. 2016.

Presentations & Talks

Visual concept-driven image generation with text-to-image diffusion model
Oral Presentation, Conference on Robots and Vision, Calgary, Canada. 2025
Make-a-story: Visual memory conditioned consistent story generation
Poster Presentation, Vector Institute for AI, Toronto, Canada. 2024.
On effective learning for multimodal data
Seminar, Vector Institute for AI, Toronto, Canada. 2024.
An improved attention for visual question answering

Virtual Presentation, CVPRW 2021.

Weakly-supervised audio-visual sound source detection and separation

Virtual Presentation, ICME 2021.

Publications

1. Jiayun Luo, Wan-Cyuan Fan, Lyuyang Wang, Xiangteng He, **Tanzila Rahman**, Purang Abolmaesumi, Leonid Sigal. To Sink or Not to Sink: Visual Information Pathways in Large Vision-Language Models, ICLR 2026.
2. Wan-Cyuan Fan, **Tanzila Rahman**, Leonid Sigal. MMFactory: A Universal Solution Search Engine for Vision-Language Tasks, 2026 (under review).
3. **Tanzila Rahman**, Shweta Mahajan, Hsin-Ying Lee, Jian Ren, Sergey Tulyakov, Leonid Sigal. Visual Concept-driven Image Generation with Text-to-Image Diffusion Model. CRV 2025. (Oral)
4. Shweta Mahajan, **Tanzila Rahman**, Kwang Moo Yi, Leonid Sigal. Prompting Hard or Hardly Prompting: Prompt Inversion for Text-to-Image Diffusion Models. CVPR 2024.
5. **Tanzila Rahman,** Hsin-Ying Lee, Jian Ren, Sergey Tulyakov, Shweta Mahajan, Leonid Sigal. Make-A-Story: Visual Memory Conditioned Consistent Story Generation, CVPR 2023.
6. **Tanzila Rahman**, Mengyu Yang and Leonid Sigal. TriBERT: Human-centric Audio-visual Representation Learning. NeurIPS 2021.
7. **Tanzila Rahman**, Shih-Han Chou, Leonid Sigal, and Giuseppe Carenini. An Improved Attention for Visual Question Answering. CVPRW 2021.
8. **Tanzila Rahman** and Leonid Sigal. Weakly-supervised Audio-visual Sound Source Detection and Separation. ICME 2021 (Oral).
9. **Tanzila Rahman,** Bicheng Xu, and Leonid Sigal. Watch, Listen, and Tell: Multi-modal Weakly Supervised Dense Event Captioning. ICCV 2019.
10. **Tanzila Rahman,** Mrigank Rochan and Yang Wang. Convolutional Temporal Attention Model for Video-based Person Re-identification. ICME 2019.
11. **Tanzila Rahman**, Mrigank Rochan and Yang Wang. Video-based Person Re-identification Using Refined Attention Networks. AVSS 2019.
12. Shivansh Rao, Peng Cao, **Tanzila Rahman**, Mrigank Rochan, and Yang Wang. Non-local Attentive Temporal Network for Video-based Person Re-Identification. AVSS 2019.
13. Md Atiqur Rahman, **Tanzila Rahman**, Robert Laganière, Noman Mohammed, and Yang Wang. Membership Inference Attack against Differentially Private Deep Model. TDP 2018.
14. **Tanzila Rahman**, Mrigank Rochan and Yang Wang. Person Re-Identification by Localizing Discriminative Regions. BMVC 2017.
15. Taimur Islam, Ataur Rahman Bappy, **Tanzila Rahman,** and Mohammad Shorif Uddin. Filtering Political Sentiment in Social Media from Textual Information. ICIEV 2016.
16. Golam Moazzam, **Tanzila Rahman,** and Mohammad Shorif Uddin. Effective Techniques for Reduction of Impulse, Gaussian, and Speckle Noises. IJCSIS 2016.
17. Liton Jude Rozario, **Tanzila Rahman,** and Mohammad Shorif Uddin. Segmentation of the Region of Defects in Fruits and Vegetables. IJCSIS 2016.
18. **Tanzila Rahman**, Liton Jude Rozario, and Mohammad Shorif Uddin. An Efficient Approach for Removal of Impulse Noise from Highly Corrupted Images. SOP TSP 2015.
19. **Tanzila Rahman**, Mohammad Reduanul Haque, Liton Jude Rozario, and Mohammad Shorif Uddin. Gaussian Noise Reduction in Digital Images using a Modified Fuzzy Filter. ICCIT 2014.
20. **Tanzila Rahman** and Md. Shorif Uddin. Removal of High-Density Impulse Noise from Color Images using an Adaptive Fuzzy Filter. ICEEICT 2014.
21. **Tanzila Rahman** and Md. Shorif Uddin. Speckle Noise Reduction and Segmentation of Kidney Regions from Ultrasound Image. ICIEV 2013.
22. Md. Shorif Uddin, Madeena Sultana, **Tanzila Rahman**, Umme Sayma Busra. Extraction of text from a scene image using a morphology-based approach. ICIEV 2012.

23. Md. Shorif Uddin, **Tanzila Rahman**, Umme Sayma Busra, and Madeena Sultana. Automated Extraction of Text from Images using Morphology-Based Approach. IJEI 2012.

Preprints

1. **Tanzila Rahman**, Renjie Liao, Leonid Sigal. All in One: A Unified Synthetic Data Pipeline for Multimodal Video Understanding. 2026.
2. Mehran Taghian, Yunke Peng, Xing Huang, Yao Wang, Yaoyuan Wang, Wei Guo, Yuanyong Luo, Tianchi Hu, Junsong Wang, Xin Wang, Hu Liu, Yu Cheng, Ziwei Yu, Hongliang Li, Mehdi Rahimifar, Lei Yan, Xuefei Wang, Zhuang Ma, Lei Liu, Hui Yu, Anandharaju Durai Raju, Hoang Le, Hei Yi Mak, **Tanzila Rahman**, Shadan Golestan. HiFloat4 Format for Language Model Pre-training on Ascend NPUs. 2026. (under review)